

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Mathematics

1. A four-digit number 7A3B is divisible by 12. What is the sum of all possible values of A + B?
 (A) 28 (B) 32
 (C) 40 (D) 38
2. A cube has volume 4913 cm^3 . If each side is increased by 2 cm, what is the new volume?
 (A) 5832 cm^3 (B) 6859 cm^3
 (C) 8000 cm^3 (D) 9261 cm^3
3. **Statement I:** Every quadratic polynomial has at most two linear factors.
Statement II: The degree of a polynomial determines the maximum number of its factors.
 (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is true and Statement II is false
 (D) Statement I is false and Statement II is true
4. The area of a square garden is 2025 m^2 . If a path of uniform width 2 m is built around it, what is the area of the path?
 (A) 256 m^2 (B) 376 m^2
 (C) 400 m^2 (D) 484 m^2
5. A tank has $\frac{3}{5}$ of its capacity filled with water. After adding 120 liters, it becomes $\frac{4}{5}$ full. What is the total capacity of the tank?
 (A) 400 L (B) 500 L
 (C) 600 L (D) 800 L
6. A car travels at $(3x+5) \text{ km/h}$ for $(2x-1)$ hours. How far does it travel?
 (A) $6x^2 + 7x - 5$ (B) $6x^2 - 7x - 5$
 (C) $6x^2 + 13x - 5$ (D) $6x^2 + 7x + 5$
7. A perfect square number has 99 as its last two digits. How many such four-digit numbers exist?
 (A) 0 (B) 1
 (C) 2 (D) 3

8. In a school, students are arranged in rows for assembly. If they are arranged in rows of 18, 24, or 30, exactly 5 students are left out each time.

What is the smallest possible number of students?

- (A) 365 (B) 720
(C) 725 (D) 900

9. Simplify: $(p + 2q)^2 - (p - 2q)^2$

- (A) $4pq$ (B) $8pq$
(C) $4p^2$ (D) $8q^2$

10. The HCF of two numbers is 12 and their LCM is 360. If one number is 60, what is the other?

- (A) 48 (B) 72
(C) 84 (D) 96

11. The speed of light is approximately 3×10^8 m/s. How long does it take for light to travel 9×10^{15} meters?

- (A) 3×10^7 seconds
(B) 3×10^6 seconds
(C) 3×10^8 seconds
(D) 3×10^9 seconds

12. Ravi's mother gave him ₹120 to buy books. He spent $\frac{3}{8}$ of it on a storybook and then $\frac{2}{5}$ of the remaining on a notebook.

How much money is left with him?

- (A) ₹36 (B) ₹45
(C) ₹54 (D) ₹60

13. If $\sqrt{x+4} + \sqrt{x-4} = 4$, what is the value of x ?

- (A) 16 (B) 5
(C) 36 (D) 49

14. A bacteria culture doubles every hour. If there are 500 bacteria initially, how many will there be after 6 hours?

- (A) 500×2^5 (B) 500×2^6
(C) 500×6^2 (D) 500×2^{12}

15. Rohan bought $(2a+3)$ notebooks at ₹ $(a-2)$ each. How much did he spend in total?

- (A) $2a^2 - a - 6$ (B) $2a^2 + a - 6$
(C) $2a^2 - a + 6$ (D) $2a^2 + a + 6$

16. A recipe requires $\frac{3}{4}$ cup of sugar for one batch. If Riya wants to make $\frac{2}{3}$ of a batch, how much sugar does she need?

- (A) $\frac{1}{2}$ cup (B) $\frac{2}{3}$ cup
(C) $\frac{3}{5}$ cup (D) $\frac{5}{8}$ cup

17. The cube of a number is 8 times the cube of another number. If the sum of the numbers is 6, what is the smaller number?

- (A) 1 (B) 2
(C) 3 (D) 4

18. In a test, Kavya answered $\frac{5}{6}$ of the questions correctly. If there were 48 questions, how many did she answer incorrectly?

- (A) 6 (B) 8
(C) 10 (D) 12

19. Which of the following is incorrectly matched?

**Polynomial
Type**

**Factorisation
Pattern**

- | | |
|------------------------------|-------------------------------------|
| (A) Difference of Squares | $a^2 - b^2 = (a+b)(a-b)$ |
| (B) Perfect Square Trinomial | $a^2 + 2ab + b^2 = (a+b)^2$ |
| (C) Sum of Cubes | $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$ |
| (D) Quadratic Trinomial | Always factorises as $(x+p)(x+q)$ |

20. If the 8-digit number 1234A678 is divisible by 11, what is the value of A?

- (A) 1 (B) 9
(C) 5 (D) 7

21. The sum of three consecutive multiples of x is $3x + 6$. What is the middle multiple?

- (A) $x + 2$ (B) $x + 1$
(C) x (D) $x - 1$

22. If $\sqrt[3]{0.000512 \times x} = 0.8$, what is the value of x ?

- (A) 100 (B) 1000
(C) 10000 (D) 100000

23. The sum of the squares of three consecutive even numbers is 440. What is the middle number?

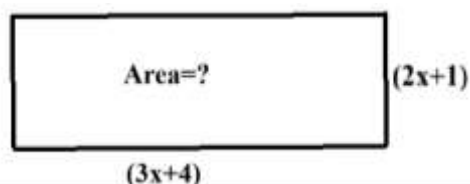
- (A) 10 (B) 12
(C) 14 (D) 16

24. In a computer system, file sizes are measured in bytes, where 1 KB

$= 2^{10}$ bytes and 1 MB $= 2^{20}$ bytes. How many KB are there in 1 MB?

- (A) 2^{10} (B) 2^{20}
(C) 2^{30} (D) 2^{40}

25. A rectangular garden has length $(3x+4)$ meters and breadth $(2x-1)$ meters. What is the area of the garden?



- (A) $6x^2 + 5x - 4$ (B) $6x^2 + 11x - 4$
(C) $6x^2 - 5x + 4$ (D) $6x^2 - 11x + 4$

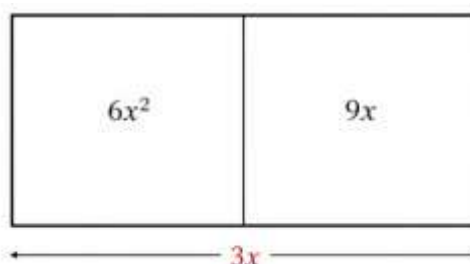
26. A square has an area of 4^3 cm^2 . What is the length of its side?

- (A) 4 cm (B) 8 cm
(C) 16 cm (D) 64 cm

27. A perfect cube number ends with 64. What is the unit digit of its cube root?

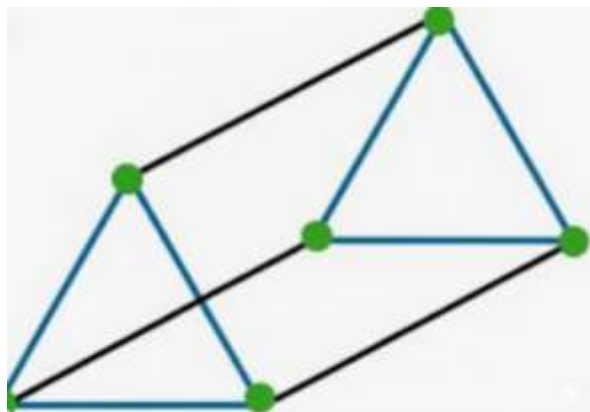
- (A) 4 (B) 6
(C) 8 (D) 2

28. The diagram below shows a rectangle split into two smaller rectangles. The total area is represented as: $\text{Area} = 6x^2 + 9x$ the length of the whole rectangle is $3x$. What is the width of the rectangle in factorised form?



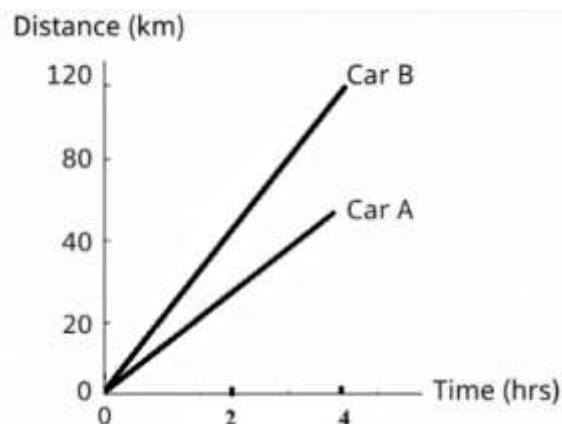
- (A) $2x + 3$ (B) $3x + 3$
(C) $x + 3$ (D) $3x + 2$

29. A polyhedron is drawn with: 6 vertices, 9 edges. How many faces does it have according to Euler's formula?



- (A) 4 (B) 5
(C) 6 (D) 7
30. A train covers a distance in 5 hours. If its speed were 10 km/h more, it would take 4 hours. What is the original speed?
(A) 40 km/h (B) 50 km/h
(C) 60 km/h (D) 70 km/h
31. A shirt priced at ₹800 is sold at a discount of 15%. What is the selling price?
(A) ₹680 (B) ₹720
(C) ₹640 (D) ₹700
32. A conical camping tent has a base radius of 7m and a vertical height of 24m. How much canvas (in square meters) is required to make the tent (excluding the floor)?
(A) $528 m^2$ (B) $550 m^2$
(C) $1232 m^2$ (D) $704 m^2$
33. A TV is marked at ₹25,000. After allowing two successive discounts of 10% and 5%, its final price is:
(A) ₹21,375 (B) ₹21,500
(C) ₹22,000 (D) ₹20,250

34. The graph below shows the distance travelled by two cars, A and B, over time.



If Car B travels at a constant speed, how far will it go in 6 hours?

- (A) 100 km (B) 180 km
(C) 140 km (D) 160 km
35. A shopkeeper packs $24xy^2$ chocolates into boxes, each containing $8y$ chocolates. How many boxes are needed?
(A) $3xy$ (B) $3x$
(C) $3y$ (D) $24xy$
36. If $\frac{3x-2}{4} = \frac{x+7}{3}$, then x is:
(A) 5 (B) 9
(C) 6.8 (D) 8
37. A cylindrical water tank has a radius of 1.4 m and height 3 m. How many litres of water can it hold? (Take $\pi = 22/7$)
(A) 18480 L (B) 1848 L
(C) 184.8 L (D) 17480 L
38. A solid has no vertex and no edge. It rolls smoothly on a flat surface. Which solid is it?
(A) Cube (B) Cylinder
(C) Sphere (D) Cone

39. Ritu's age is twice that of her sister. In 5 years, the sum of their ages will be 31. What is Ritu's present age?

- (A) 12 (B) 14
(C) 16 (D) 18

40. If 12 workers can build a wall in 9 days, how many workers are needed to build it in 6 days?

- (A) 15 (B) 18
(C) 20 (D) 24

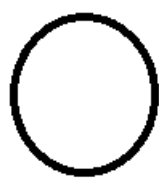
41. The walls and ceiling of a room measuring $8\text{ m} \times 6\text{ m} \times 4\text{ m}$ are to be painted. The room has a door of size $2\text{ m} \times 1\text{ m}$ and two windows each $1.5\text{ m} \times 1\text{ m}$. If painting costs ₹25 per m^2 , what is the total cost?

- (A) ₹3875 (B) ₹4250
(C) ₹4850 (D) ₹5350

42. After spending 25% of her money, Priya has ₹450 left. How much did she have initially?

- (A) ₹600 (B) ₹550
(C) ₹500 (D) ₹650

43. Which solid matches these views?



Top view



Side View

- (A) Cylinder (B) Cone
(C) Sphere (D) Triangular prism

44. A car travels 240 km on 20 litres of petrol. How far can it travel on 35 litres under the same conditions?

- (A) 400 km (B) 420 km
(C) 380 km (D) 360 km

45. In a polynomial $3a^2b - 6ab^2$, the greatest common factor is:

- (A) $3ab$ (B) $6a^2b^2$
(C) ab (D) $3a^2b$

46. A trapezium-shaped field has parallel sides 40 m and 60 m, and the distance between them is 25 m. What is the cost of levelling it at ₹5 per m^2 ?

- (A) ₹6250 (B) ₹7500
(C) ₹8750 (D) ₹10000

47. The simple interest on ₹5000 at 8% per annum for 3 years is:

- (A) ₹1000 (B) ₹1200
(C) ₹1400 (D) ₹1600

48. In a dice, the number of dots on opposite faces always adds up to 7. If the front face shows 5 dots, what number is on the back face?

- (A) 1 (B) 2
(C) 6 (D) 3

49. A cone and a cylinder have the same base radius and height. The ratio of the volume of the cone to that of the cylinder is:

- (A) 1:1 (B) 1:2
(C) 1:3 (D) 2:3

50. A box measures $12\text{ cm} \times 8\text{ cm} \times 5\text{ cm}$. What is the volume of the box?

- (A) 480 cm^3 (B) 520 cm^3
(C) 600 cm^3 (D) 720 cm^3

